**LOURDES MATHA COLLEGE OF SCIENCE AND TECHNOLOGY**

**DEPARTMENT OF COMPUTER APPLICATIONS**

**Semester: 3rd** **Branch: MCA** **Admission Year: 2021**

**First Series Examination / October 2022**

**20MCA283 /DEEP LEARNING**

**Time:1 ½ Hours** **Max. Marks: 30**

## Part-A (Answer All Questions)

| Q.No | Questions | Marks | CO | BL |
|---|---|---|---|---|
| 1 | Describe sigmoid activation functions. | 3 | CO1 | L2 |
| 2 | Write the gradient descent algorithm. | 3 | CO1 | L2 |
| 3 | An input image has been converted into a matrix of size 12 X 12 along with a filter of size 3 X 3 with a Stride of 1. Determine the size of the convoluted matrix. | 3 | CO3 | L3 |
| 4 | Explain perceptron learning algorithm. | 3 | CO1 | L2 |
| 5 | Explain the ideas of Rank, Shape and Type with an example in the context of a Tensor Data Structure | 3 | CO1 | L2 |
| 6. | Describe the model of a biological neuron. | 3 | CO1 | L2 |
| | **Part-B (Answer all Questions)** | | | |
| 7 | With a suitable example explain how back propagation works. | 6 | CO1 | L2 |
| 8 | Explain the role of batch normalization in training a neural network and describe how to find out over fitting from training and validation curves. | 6 | CO1 | L2 |